

TECHNICAL DATA SHEET

BRSCMC — Sodium Carboxymethyl Cellulose (CMC), Oil Drilling Grade

Viscosifier, Flow-Control Agent and Filtration Reducer for Drilling and Completion Fluids

1. Product Description

Product names: Carboxymethylcellulose, CMC, CMC Na, Sodium Carboxymethylcellulose, Cellulose Gum, Sodium CMC

BRSCMC is an anionic cellulose ether with a macromolecular structure. It is an odorless, tasteless, and non-toxic white or slightly yellowish free-flowing powder or granule. It dissolves readily in water to form a colloidal solution and is insoluble in ethanol, ether, acetone, and other organic solvents.

2. Main Functions

In drilling fluids, BRSCMC acts as a viscosifier, flow-control agent, and filtration reducer. It provides strong cuttings-suspension and carrying capacity, cleans cuttings from the borehole, stabilizes the wellbore, protects the drill bit, and forms a thin mud cake. In completion fluids, it controls viscosity, suspends weighting materials and fillers, and prevents fluid loss. It can also replace guar gum in fracturing fluids.

3. Features and Advantages

- Low dosage with high yield
- Good salt tolerance and antibacterial resistance
- Combines fluid-loss reduction with viscosity improvement
- Strong suspending and flow-control capability
- Environmentally friendly, non-toxic, harmless, and odorless
- Good fluidity — convenient to handle and apply



4. Typical Applications

Application	Key Benefits
Non-dispersed drilling fluids	Carries cuttings, inhibits clay dispersion, stabilizes the wellbore, and increases drilling rate
Dispersed drilling fluids	Strong solids suspension; ideal for high-density fluids; forms a thin, compact mud cake and reduces fluid loss and free water
Calcium-treated fluids	Excellent calcium resistance; prevents excessive flocculation of clay and maintains stable rheology
Salt-water, seawater & saturated brine fluids	Low salt sensitivity; fast rheology adjustment at low dosage; filtrate close to native formation water, minimizing damage to the pay zone
Potassium-based fluids	Low sensitivity to potassium, calcium, and magnesium salts; efficiently adjusts rheology and cleans cuttings and the bit
Polymer & low-solids fluids	Good polymer compatibility; maintains low solids and low clay dispersion; effective filtration control

Application	Key Benefits
Environmentally sensitive operations	Non-toxic, odorless, and biodegradable; low fluid-maintenance cost; no harm to agricultural production
Completion & packer fluids	Improves slurry fluidity with low filtration loss; protects the wellbore; highly effective in clean-brine packer fluids owing to good monovalent-salt tolerance
Fracturing fluids	Builds viscosity quickly, carries proppant into fractures, and reduces fluid loss; residue-free and can replace guar gum
Granulated & instant grades	Dust-free and fast-dissolving; rapid dispersion even under rugged field conditions

5. Packaging and Dosage

Packaging: 25 kg triple-layer paper sacks with a PE (polyethylene) inner liner, or according to customer requirements

Typical dosage: 0.2% – 1.0%

6. Storage and Shelf Life

Conditions: Store in a cool, dry, clean, and well-ventilated area at 10–30 °C (maximum 40 °C) and relative humidity ≤ 80% (ideally < 50%)

Protection: CMC is hygroscopic — protect from moisture, rain, direct sunlight, heat sources, and open flames. Keep away from strong oxidizing agents, strong acids, and strong alkalis

After opening: Expel excess air and reseal the inner bag immediately; repack damaged sacks into sealed, moisture-resistant containers

Shelf life: Two (2) years from the date of manufacture

7. Supplier Information

Manufacturer / Supplier: Brighter Star Drilling Fluids

Website: www.pcenigeria.com

8. Disclaimer

All data, recommendations, and information contained in this Technical Data Sheet are provided for reference only. They are derived from laboratory testing and field experience obtained under specific conditions, and actual product performance may vary with formation characteristics, water quality, equipment, and operating practices. Users are advised to verify the suitability of this product through on-site trials and to adjust the dosage according to actual working conditions. In all cases, the actual conditions and results of use shall prevail.

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